

Blockchain-Based Carbon Footprint Management

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Abstract. This paper introduces a novel approach to managing carbon footprints using blockchain technology to integrate these footprints intrinsically into the attributes of products, akin to their price. In contrast to conventional methods that treat carbon footprints as distinct, tradeable units, our model incorporates them directly into the product life cycle, thus maintaining the connection between environmental impact and product consumption. By closely examining blockchain’s functionality, this study illustrates how it can improve transparency and security in transactions while influencing market dynamics by making carbon accountability an integral part of product ownership. This method may aid in establishing a more sustainable economic model by better incorporating environmental costs into transactions, potentially fostering progress in environmental finance and sustainability initiatives.

Keywords: Blockchain · CO2 · Blockchain Technology · CO2 Emissions · Carbon Footprint Management · Carbon Credits · Environmental Sustainability · Tokenization · Token Economy · Tokenization of Assets · Utility Tokens · Multiple Attribute System · Smart Contracts for Data Sharing.

1 Introduction and Background

1.1 Motivation

Currently, the economic system operates primarily on the basis of the monetary price alone. In contrast, it is conceivable to tokenize each attribute of a product, including its price, and embed these attributes distinctly within the product’s characteristics, which can then be administered through blockchain technology. In particular, the carbon footprint emerges as a crucial attribute suitable for tokenization and embedding, underscoring its prominence in contemporary environmental discourse. Recognizing carbon credits as essential to environmental sustainability, this document addresses the pressing challenge of integrating carbon accountability into routine economic operations by redefining carbon credits as both additional and embedded intrinsic attributes of products and services, similar to, but distinct from, their pricing. Using blockchain technology, this innovative strategy not only challenges conventional views on environmental accountability but also harmonizes sustainability with economic activities. Traditionally, carbon has been treated as an independent commodity, disconnected

from its product of origin, and exchanged independently throughout its life cycle. This perspective commodifies carbon as an asset that can be purchased and monetized, overlooking the necessity to influence consumer behavior or production processes directly. By embedding carbon within the product’s attributes itself—similar to its price—this paper advocates for a paradigm shift. Blockchain technology has the potential to improve transparency, traceability, and equitable accountability, potentially influencing economic and environmental paradigms. Unlike existing models, this approach incorporates carbon credits directly as product attributes, thereby revamping both the economic and environmental paradigms.

The contributions of this paper are delineated in four principal areas. First, reconceptualization of carbon credits as distinct intrinsic attributes of products and services, establishing a direct association between carbon accountability and product value, thus fundamentally reshaping existing economic and environmental paradigms. Secondly, the proposition of a blockchain-based architecture that guarantees transparency, confidentiality, and tamper-resistant record keeping, thus addressing the limitations inherent in traditional systems. In addition, smart contracts are utilized for data sharing, offering a practical and effective alternative to conventional cryptographic methods of data exchange. Finally, the evaluation of the suitability of existing blockchain platforms for the deployment of this system explains the strengths and limitations of various technologies, thus providing a foundation for selecting solutions that align with a multi-attribute economic system. Specifically, the proposed blockchain-based architecture incorporates features such as smart contracts for automation and tokenization for integration.

A multi-user, privacy-preserving tamper-proof database is essential for data governance and audit trails. The key role of blockchains in ensuring tamper-proof operations and privacy is highlighted in [1]. It supports data sharing with multiple users, as noted in [2]. Blockchain technology provides tamper-proof data management, transparent governance, and privacy mechanisms, confirmed by [1], [2]. The search for encrypted records is crucial, as shown in [3]. The multistage processes of blockchain databases add versatility and complexity, as discussed in [4]. This combination significantly advances database technology, meeting modern data governance needs. The secure and decentralized nature of blockchain supports a viable economic model for tracking carbon credits in products and services.

1.2 Background

Our model [12] assigns both the price and tokenized carbon footprint to the products, challenging traditional economic paradigms that historically have lacked the capacity to manage multiple attributes per product. Blockchain technology addresses this limitation by facilitating the integration and tracking of numerous attributes, thereby presenting a transformative potential for conventional economic systems. We contend that carbon accountability should be considered as fundamental as pricing in fostering conscientious consumption. Existing

strategies perceive carbon credits as mere tradable commodities, thereby detaching them from their origins and contributing to a superficial market that inadequately modifies behavioral or production norms. This often disproportionately burdens those with fewer financial resources, exacerbating inequalities. Our model incorporates carbon accountability as a core attribute, aiming to promote sustainability as an important value, moving beyond its commodification. This approach fosters an equitable economic framework in which carbon costs are equitably distributed, thus promoting sustainable practices. The integration of environmental considerations with economic practices is pivotal in ensuring that the true ecological costs are accurately represented, thus cultivating a market that is informed and responsible. Blockchain can be used to improve transparency and accountability by integrating carbon credits as fundamental attributes, with the aim of better aligning environmental stewardship with market behavior. This initiative aims to support more sustainable consumption and align with ethical standards, contributing to the development of a more just and sustainable economic framework.

1.3 Related Work

The article by Kalaiselvan et al. [7] introduces a blockchain-based system to transform carbon credit markets by addressing inefficiencies such as overcrediting and high costs. It proposes a decentralized marketplace to improve transparency, reduce expenses, and improve collaboration among stakeholders. The platform provides a secure and accessible environment for trading carbon credits, using a decentralized ledger to ensure transaction transparency and prevent fraud. Smart contracts automate processes, lowering costs, and supporting stakeholder collaboration. The integration of blockchain is intended to streamline trading, improve regulatory compliance, and promote sustainable practices. The initiative emphasizes a strong governance structure that involves diverse stakeholders for collective decision making, aligning with global efforts to reduce greenhouse gas emissions. It aims to serve as a model for future carbon trading platforms through technological innovation. The research by Jawalkar et al. [8] proposes a blockchain framework to tokenize and securely transfer carbon credits, in order to improve accessibility and market efficiency. Addresses issues in traditional carbon credit systems—such as fragmentation, lack of transparency, high costs, and overcrediting—by establishing a Carbon Credit Ecosystem. This system uses smart contracts to improve market consistency, liquidity, and transparency, with key features such as fair token distribution and an automated market maker for efficient trading. The project seeks to reduce fraud and errors through secure digitalization, promoting inclusivity, transparency, and efficiency in carbon trading and other systems to combat climate change. Implementation relies on smart contracts and blockchain for task automation, transaction transparency and security, and verification of emission reduction. The study by Zhang et al. [9] explores how blockchain technology can enhance traceability in low carbon supply chains using evolutionary game theory. It highlights the broader potential of blockchain beyond carbon credits, particularly in managing carbon emissions

within supply chains. The research focuses on the influence of government policies and stakeholder relationships, employing mathematical modeling and MATLAB simulations to identify optimal blockchain adoption strategies for manufacturers, suppliers, and governments. Key findings suggest that manufacturers and suppliers should manage blockchain costs while collaborating on eco-friendly certifications. Local governments are encouraged to foster stakeholder participation through reward and punishment systems. In general, blockchain has been shown to improve transparency and accuracy in tracking carbon footprints while reducing costs, with collaborative strategies and governmental support crucial to successful technology adoption. The article by Swinkels [10] examines the role of blockchain in carbon credit trading, focusing on tokenization, pricing, and market dynamics. It highlights the efficiency improvements blockchain provides, using the ACX platform as a case study, which tokenized 3.8 million tons of CO₂e, with 2.8 million burned and 1 million available for trading by the end of 2022. Swinkels reports a turnover of USD 21.2 million in the secondary market during 2021-2022, indicating limited liquidity and pricing alignment with similar projects. The study illustrates blockchain's ability to reduce costs and enhance transparency in the carbon credit market, concluding that blockchain can significantly improve transaction processes and has wider potential in environmental finance, suggesting further exploration of its global market impact. The paper by Akiladevi et al. [11] presents a strategy using multichain blockchain technology for tokenization of energy assets, emphasizing environmental sustainability. Examines the role of blockchain in improving carbon credit markets and energy asset management through a secure decentralized ledger that improves transparency, trust, and reduces fraud. By tokenizing assets as digital tokens, the method facilitates easier trading and management, encouraging investment in renewable energy projects. Blockchain resistance to tampering, automation capabilities, and cost efficiency contribute to reducing carbon footprints and support sustainable energy development. The paper concludes by highlighting the transformative potential of multi-chain blockchain in energy management and suggests further research to address technical issues.

As corroborated by the reviewed studies, the prevalent methodology in carbon management frequently entails disassociating a product's carbon footprint from the product itself, considering it as a tradable commodity. In contrast, the approach delineated herein incorporates the carbon footprint as an integral characteristic of the products, analogous to their pricing. This paradigm establishes a direct correlation between emissions responsibility and product consumption, thereby ensuring that environmental responsibility cannot be transferred or alleviated through the acquisition of credits. Such a framework effectively deters the perpetuation of unsustainable operations by economically advantaged entities, while simultaneously addressing the disproportionate adversities encountered by less affluent individuals who contribute minimally to global emissions. By embedding the carbon footprint within the fundamental attributes of products, this model cultivates authentic sustainability and endorses equitable environmental responsibility across all socioeconomic strata.

2 Challenges in Data Management Approaches

Traditional databases lack transparency, tamper-proof security, and scalability, but the blockchain addresses these issues through decentralized architecture, ensuring data integrity and smooth integration of records such as carbon credits. The current data infrastructure is characterized by the widespread utilization of centralized database systems by organizations. These systems are conducive to facilitating audits; however, they hinder cross-industry enhancement due to data siloing, especially within complex logistics and supply chains. Centralized databases, while effective in optimizing data management processes, frequently contribute to an increased carbon footprint. The current auditing methodologies employed by government and centralized systems, which utilize central servers, necessitate a re-evaluation to improve environmental sustainability. Analogous Applications across Industries: Homogeneous database systems that span various sectors present both environmental challenges and opportunities in terms of the management of carbon footprints. Decentralized Data Management: Isolated databases hinder collaboration and optimization in logistics and supply chains, increasing carbon footprints through inefficiencies. Isolated systems hinder efficiency and sustainability, complicating carbon footprint reduction efforts across industries.

2.1 Existing Systems for Data Collection

The Point of Sale (POS) system, as shown in *Figure 1* is essential for modern business operations and streamlined retail transactions. Invoicing: Records transactions to ensure sales transparency and accuracy, helping businesses and customers understand purchases. Stock/Inventory Management: Manage product stock, improve restocking efficiency, minimize waste, and improve operations. Customer Loyalty Programs: POS programs also handle customer rewards and promotions, promoting retention and lasting engagement. (as demonstrated in *Figure 2*)

Transaction data collection at critical checkpoints is managed by a Data Acquisition System in devices like 'Cash Registers' or 'POS Terminals.' These checkpoints efficiently capture transactional data, which is a key part of the data lifecycle. Data Preservation Methods: Transaction records are stored in plain text within protected memory, or in sealed memory chips such as EEPROMs to ensure accessibility and integrity, or encrypted for critical audits to secure sensitive information. Navigating Data Management Challenges: Organizations balance data security, privacy, and collaboration in data management. Enhancing multi-user capabilities and governance structures transforms data infrastructure. Cryptographic protocols such as "Public Key Encryption with Keyword Search" simplify database use and sharing by managing data access permissions more effectively, as noted in [5].

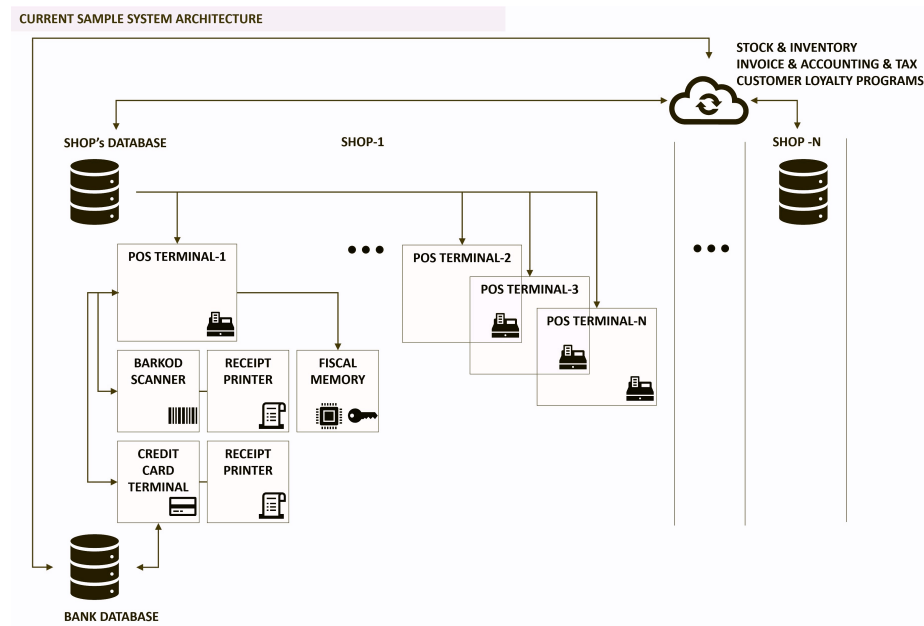


Fig. 1. Current Sample System Architecture

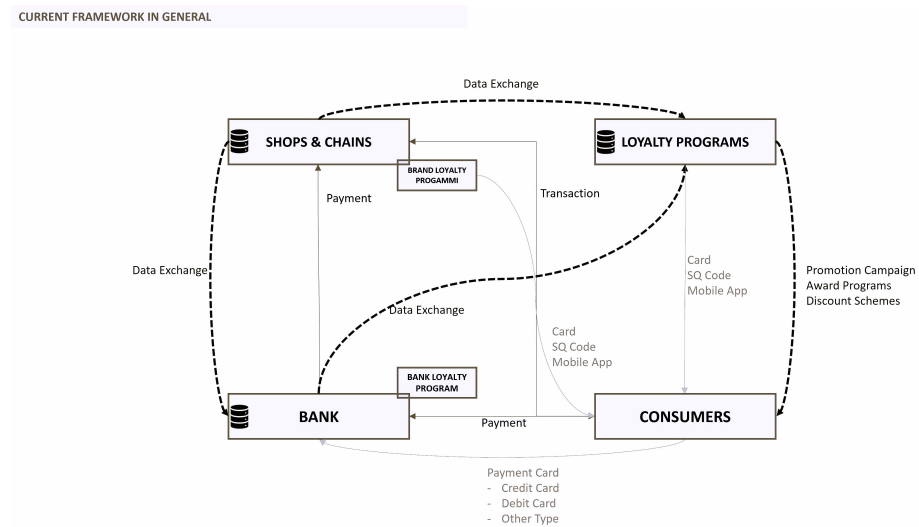


Fig. 2. Current Framework in General

2.2 What is Missing in Current Systems

In spite of progress in digital infrastructure, present systems possess significant deficiencies impacting efficiency, security, and privacy. These challenges are predominantly: firstly, challenges in data sharing, which arise from incompatible formats and platforms. Secondly, there are common database management difficulties, where maintaining data integrity and security during database management is strenuous. In addition, there exists the problem of data silos, which obstruct efficient utilization and integration. Furthermore, designing systems for multiple users introduces complications, as these systems must navigate access permission issues, striking a balance between accessibility and security. Lastly, privacy concerns in the management of sensitive records, particularly those related to healthcare, require a high degree of confidentiality. Recognizing these concerns underscores the necessity for blockchain technology to provide decentralized, secure, and transparent data management. To address this, an approach involves the implementation of end-to-end encryption, which ensures that each transaction within the database is safeguarded, maintaining the security of data from its creation to its ultimate destination. This encryption is pivotal in protecting sensitive information against unauthorized access and potential security breaches. Another approach emphasizes efficient keyword search over encrypted data; this crucial feature allows conducting efficient keyword searches on encrypted information, thus resolving the typical trade-off between data security and usability and enabling users to query encrypted data without privacy compromise. The research carried out by Jiang et al. [6] supports the feasibility of this solution, demonstrating how blockchain technology can be used effectively to create a secure and privacy-preserving database. Their findings highlight the potential for blockchain-based systems to revolutionize data management practices, particularly when it comes to addressing privacy and security concerns.

2.3 The Need for Blockchain

Blockchain databases, by employing cryptographic methods and encryption, offer a novel approach to data management that can improve the tracking of carbon footprints and potentially enhance security, transparency, and overall efficiency. Firstly, blockchain implementation makes tokenization feasible. Secondly, blockchain ensures exceptional immutability and security, with cryptographic chaining fostering tamper-evident records that uphold data integrity and provide a verifiable history. In addition, innovative record sharing is facilitated through searchable attribute-based encryption (ABE), allowing data owners to encrypt records with attributes, allowing authorized entities to access precise and secure data while preserving privacy. Furthermore, the blockchain fosters decentralized trust by eliminating central authorities and using consensus-driven validations for democratic and transparent data management. Furthermore, smart contracts within blockchain systems automate complex processes and transactions based on predefined rules, thereby reducing the reliance on intermediaries. Finally, the shared blockchain database supports multi-stakeholder collaboration, leading to

optimized operations, improved data sharing, and reduced redundancies. In summary, blockchain databases advance data management by offering solutions for immutability, record sharing, decentralized governance, and automation, thus establishing a secure, transparent, and efficient digital infrastructure for data storage, sharing, and use across various industries.

2.4 Blockchain-Based Database for Environmental and Carbon Footprint Management

In addition to tokenization, implementing a blockchain-based database for environmental management improves data handling by allowing multi-party transaction recording. This advancement allows data sharing, improves integrity, and usability without the need for replication. The primary benefits include the following: Firstly, *No Replication of Data* eliminates the redundancy found in traditional systems, ensuring that data are singular and authoritative. This process reduces inconsistencies, streamlining management for both efficiency and versatility. Secondly, *Versatility for Other Purposes* means that while it is primarily used for environmental data, it can be extended to various applications, such as marketing and delivery optimization, transforming consolidated data into actionable insights. Moreover, *Multi-Stakeholder and Sector-Wide Optimization* is achieved through its user-friendly design that fosters collaboration among stakeholders, optimizing operations both at the organizational level and across sectors. In addition, *Data Exchange Between Industry Layers* is facilitated by blockchain, which enables smooth data transfer between industry layers, creating a more integrated and responsive business ecosystem.

The blockchain-based database enhances carbon footprint management through its robust security and transparency features. It offers a secure, efficient framework for handling carbon data, ensuring data integrity and confidentiality via cryptography. To begin with, secure transaction management involves encrypting transactions to safeguard environmental data against unauthorized access. Next, the system allows effective data management and querying without jeopardizing integrity. Moreover, integrating smart contracts automates transactions, streamlining operations, and boosting efficiency and transparency.

3 Methodology

Enhancing the carbon footprint tracking system with additional features improves the accuracy of the data and its integration into existing processes. Embedding data collection and carbon credit payment terminals in retail stores, similar to POS terminals, and using blockchain technology can transform transaction methods. This not only simplifies carbon credit payments, but also ensures transparency, such as credit card transactions. Blockchain integration offers immutability, transparency, and trust, ensuring carbon data and transactions are secure and reliable, fostering trust among consumers and businesses.

3.1 Use-Case Scenario

Consider a consumer who purchases a product from a retailer that participates in the blockchain-based carbon credit system. Each product is tagged with its carbon footprint, which is verified and recorded on the blockchain. During purchase, the system automatically deducts associated carbon credits from the consumer's balance and logs the transaction immutably on the blockchain as shown in *Figure 3*. For example, a smartphone with a carbon footprint of 50 kg CO₂e requires 50 carbon credits. When the consumer completes the purchase, the system deducts 50 credits from their account and updates the blockchain ledger, ensuring transparency and accountability. This process not only provides consumers with clear visibility into the environmental impact of their purchases but also incentivizes manufacturers to adopt sustainable practices by making carbon emissions a tangible cost of production.

3.2 Proposed System Architecture

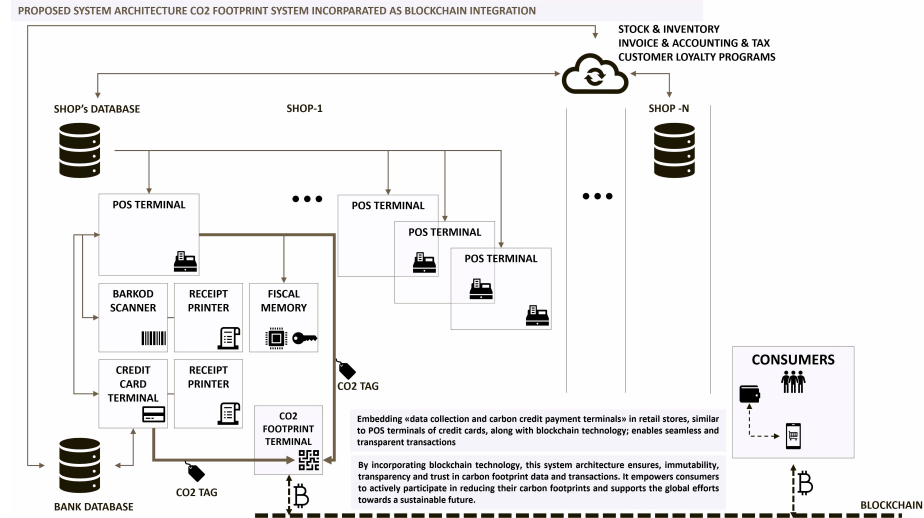
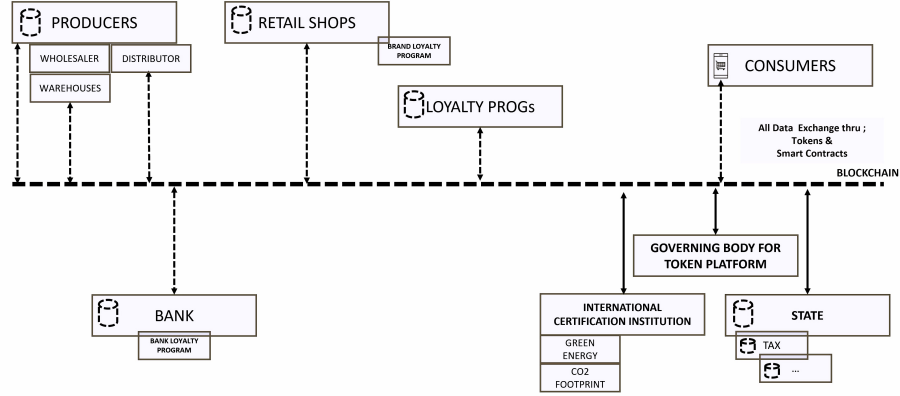


Fig. 3. Proposed System Architecture Co₂ Footprint System Incorporated As Blockchain Integration.

(The large 'B' symbol with vertical stripes in this figure is inspired by the Bitcoin logo but is used here to represent cryptocurrencies or blockchain tokens in a general sense, rather than specifically Bitcoin.)

The blockchain-based carbon credits system ensures transparency and accountability by integrating them into the lifecycle of products (*Figure 4*). Key features include smart contracts for secure transactions and a decentralized

PROPOSED SYSTEM FRAMEWORK IN GENERAL

**Fig. 4.** Proposed System Framework In General

ledger to track carbon footprints across supply chains. Scalability is enhanced through side chains or Layer-2 solutions (*where possible*), reducing costs and increasing speed. Tokens represent carbon emissions, facilitating exchanges and environmental actions, while smart contracts manage these transactions. This framework provides a decentralized, immutable, and transparent integration of carbon credit management into global supply chains.

3.3 Challenges and Limitations

The proposed system introduces the integration of carbon credits into products and services in an innovative way, yet encounters challenges during implementation. To begin with, there is the issue of scalability; blockchain platforms often contend with scalability limitations, processing only a restricted number of transactions per second, posing a challenge for sectors with high transaction volumes such as retail and manufacturing. The mitigation proposed involves adopting Layer-2 solutions, like rollups or sidechains, which could relieve scalability issues by diverting transactions away from the main chain. On the subject of blockchain's energy consumption, especially under Proof-of-Work (PoW) protocols, this presents a sustainability concern, and transitioning to more energy-efficient models such as Proof-of-Stake (PoS) is not without complexity. As a mitigation strategy, it is suggested to promote the adoption of PoS-based blockchains or private blockchain configurations that require less energy. Blockchain systems can also be cost prohibitive in terms of infrastructure and maintenance for small and medium enterprises (SMEs). Therefore, a proposed mitigation is financial assistance or subsidies from governments or NGOs, designed to defray these costs for SMEs, thus fostering broader adoption.

Additionally, data privacy and security lead to concerns: while blockchain ensures the immutability of data, it poses risks to the privacy of sensitive data,

such as proprietary carbon footprint calculations. Incorporating searchable encryption (SE) and zero-knowledge proofs adds to the complexity. Instead of relying on traditional techniques, we could adopt privacy-preserving technologies, like homomorphic encryption, enabling interaction with blockchain-based databases via smart contracts; this approach is more straightforward, demands less computational power, and facilitates economical data-sharing transactions. SE allows secure queries of encrypted carbon data, safeguarding privacy, which is essential for manufacturers. Despite SE's advantages in data privacy, its implementation on large databases can result in computational complexity. Traditional SE involves cryptographic operations that are resource-intensive and inefficient for large-scale systems; however, the more straightforward and efficient SE solution of blockchain differs. The blockchain retains encrypted data that engages with smart contracts, thereby avoiding direct data exposure. Smart contract systems possess different identifiers across blockchains, being referred to as "smart contracts" in Ethereum. A blockchain-based carbon credit system stores encrypted emissions data securely, enabling users to verify carbon footprints via smart contracts without revealing data, hence ensuring privacy and efficiency. This approach reduces encryption/decryption costs and incorporates economic dimensions, as users compensate for query transactions using tokens or cryptocurrency, guaranteeing transparency and fairness. Regulators are provided with privileged encrypted data access for audit functions, maintaining compliance. This model exploits blockchain and smart contracts to simplify SE, enhance privacy, support transparency, drive efficiency, and sustain economic viability.

Moreover, there are regulation and standardization issues: the global adoption of blockchain-based carbon credit systems encounters regulatory challenges due to disparate environmental policies and data governance standards from country to country. Mitigation proposed includes collaboration with international organizations to establish standardized frameworks for carbon credit accounting and blockchain interoperability. Finally, user adoption issues arise: resistance from stakeholders and a lack of technical expertise, including among manufacturers and consumers, may disrupt the system's adoption. A proposed mitigation for this involves educational campaigns and the development of user-friendly interfaces to foster acceptance of the proposed system.

4 Details of System Structure and Roles of Main Stakeholders

For effective management, stakeholder roles such as producers, consumers, governing bodies, and environmental organizations must be defined. The system manages carbon footprints through smart contracts and blockchain tokens, serving as an immutable ledger for carbon transactions. Users can monitor their carbon footprint and token balance through digital wallets. The system rewards eco-friendly actions such as reforestation and support of renewable energy with tokens. Blockchain ensures reliability and transparency, fostering sustainable

consumer participation. Users earn tokens for verified green activities, such as tree planting and recycling. Validators, like environmental organizations, verify these activities and distribute rewards, promoting sustainable behaviors. Carbon tracking and offset use blockchain for accurate transaction records, giving users transparency in their carbon use and tokens, enhancing sustainability responsibility. The system links producers and consumers with smart contracts and tokens, ensuring secure exchanges and promoting sustainable practices. Here is the list of main stakeholders with brief role descriptions and functions.

1.Producers The system relies on producers to establish the carbon footprint of the product according to the guidelines of the International Certification Institute (ICI). Assesses the use of green energy in the sources, manufacturing, packaging, and distribution to provide precise environmental impact data. Manufacturers calculate carbon footprints using the ICI criteria, focusing on integrating green energy in the extraction, manufacturing, packaging, and distribution of raw materials.

2.International Certification Institution The International Certification Institute (ICI) establishes standards for carbon footprints, devises guidelines for their computation, and provides blockchain-based tokens to measure carbon emissions. Firstly, ICI defines and standardizes carbon footprints. Secondly, it develops guidelines and methodologies for their calculation. In addition, factors such as production methods, materials, energy consumption, and transportation are taken into account. Then, ICI issues carbon footprint tokens, followed by generating and securely storing these tokens on a transparent blockchain platform.

3.The State The proposed framework employs blockchain technology to incentivize environmentally sustainable behaviors by issuing carbon tokens that are authenticated by environmental organizations. In addition, the system facilitates carbon tracking and fosters sustainability education.

4.The Governing Body The regulatory authority enhances the operational capacity of the carbon footprint token ecosystem by overseeing token issuance aligned with carbon emissions, ensuring a consistent supply, and modifying conversion rates to prevent inflation. By partnering with carbon offset initiatives, it aids in emissions reduction, while blockchain technology provides security and transparency in transactions. Additionally, it fosters awareness through educational programs and collaborates with regulators to uphold the system's integrity. The main functions include: firstly, token issuance; secondly, regulating token supply and distribution; next, adjusting the token conversion rate; furthermore, overseeing token redemption and offset; in addition, ensuring token utility and rewards; also, managing token burning; moreover, facilitating carbon credit trading; besides, maintaining transparency and auditing; additionally, advancing education and awareness; and finally, ensuring regulatory compliance.

5. Consumers In the proposed framework, consumers play an essential role, as their choices in buying and participating in environmental practices heavily influence the success of carbon footprint management. Within this system, consumers play a key role in financial and purchasing transactions. When a product

is bought, consumers pay not only the item’s price but also the related carbon footprint cost through a digital wallet or payment platform linked to a blockchain network. This setup automatically deducts the necessary carbon footprint tokens from the consumer’s wallet, thus aiding in the tracking and reduction of carbon emissions. To elaborate on these processes, First, environmental activities are verified, after which token rewards are distributed. Following this, the redemption and offset of carbon credits take place. In addition to this, consumers are involved in purchasing and payment activities. Furthermore, the carbon footprints of products are calculated. Alongside these, tracking and offset of carbon footprints is conducted. As a result, carbon footprint tokens are created. Finally, the blockchain platform is implemented.

6. Retail stores Retail stores serve as essential intermediaries, embedding carbon credit transactions within everyday consumer engagements, and encouraging eco-friendly practices. They employ systems similar to POS to collect data and assess the carbon footprint for each transaction. Carbon credit payment systems allow customers to use digital wallets for carbon offset activities. Stores engage consumers in recognized activities such as tree planting and recycling, which are certified by environmental organizations that distribute tokens and promote stewardship. Using blockchain technology, product costs and carbon footprints are calculated simultaneously, ensuring secure and transparent recordkeeping. Retailers provide real-time reporting, enhancing system comprehension and enabling stakeholders to produce comprehensive reports, thereby increasing accountability and raising environmental awareness. Initially, data collection and the calculation of the carbon footprint are carried out. Following this, carbon credit payment terminals are utilized. Subsequently, environmental activities are validated, and token rewards are distributed. Then, consumer purchases and payments take place. The blockchain integration is then used for secure record-keeping. Finally, reporting and transparency are ensured.

5 Suitable Blockchains

The selection of a blockchain platform for applications related to carbon footprint tracking and offset is crucial to determine the efficiency, security, and scalability of the system. To achieve interoperability and global-level scalability for the carbon footprint application, a fungible Utility Token on a public blockchain is necessary. Different blockchains come with varying features, and it is important to carefully match these attributes with the project’s specific needs. In the following, we present a concise overview of several notable blockchain options and considerations for their applicability in carbon footprint management. Among the diverse range of blockchain platforms that could be effectively harnessed to manage carbon footprints, Ethereum stands out as a notable option. Ethereum’s transition to a Proof-of-Stake (PoS) consensus mechanism aligns well with global sustainability goals by reducing energy consumption. However, the platform is challenged by substantial transaction fees, which may hinder broad adoption. In contrast, Hyperledger Fabric’s permissioned architecture facilitates data

confidentiality, rendering it a compelling choice for enterprises seeking secure blockchain applications. The prospective development of hybrid models might effectively synergize these strengths, offering improvements in scalability and privacy without sacrificing energy efficiency. Initially considered is Ethereum, followed by Hyperledger Fabric. Beyond these, other suitable candidates include Algorand, Polkadot, Tezos, and Avalanche. Additional viable alternatives are Cardano, Stellar, and EOS, as well as emerging blockchains like Hedera Hashgraph, Neo, VeChain, Nano, and Solana. Each of these platforms has unique features that make them potentially favorable for certain elements of applications aimed at tracking and offset of carbon footprint, depending on specific project requirements such as transaction speed, scalability, energy efficiency, and compliance with regulatory standards.

In the rapidly changing realm of blockchain technology, Ethereum is noteworthy due to its established ecosystem and strong smart contract capabilities. However, challenges such as scalability and transaction fees during high-demand periods must be carefully evaluated. With the shift to Proof of Stake (PoS), Ethereum considerably decreases its energy consumption, which aligns with environmental objectives; nevertheless, its transaction throughput, ranging from 15 to 30 TPS, can restrict scalability in large-scale systems.

Hyperledger Fabric, particularly suited for enterprise applications, prioritizes privacy and scalability through its permissioned network and modular consensus protocols. Although appropriate for permissioned settings, it may pose a more intricate setup challenge compared to public blockchains. Additionally, Hyperledger Fabric may not be the best fit for interstate or global-level applications.

Algorand is celebrated for its high throughput and rapid transaction completion, utilizing a pure proof-of-stake model for enhanced energy efficiency. However, its relative novelty within the blockchain sector and the evolving ecosystem support are aspects worth considering.

Polkadot emphasizes interoperability and the provision of specialized functionality in parachains, offering scalability and customization; however, its complexity and the need for a well-defined ecosystem could influence the ease of adoption.

Tezos, featuring on-chain governance and formal verification, improves security and allows stakeholders to participate in protocol upgrades. However, its ecosystem might not be as comprehensive as those of more established platforms.

Avalanche, distinguished by its subsecond finality, provides a highly scalable platform with an innovative consensus mechanism. Its approach to achieve decentralization and high throughput differentiates it, making it ideal for applications demanding rigorous performance requirements.

5.1 Blockchain Implementation Analysis

To assess the operational costs related to the issuance of sample tokens, which is essential to evaluate the economic viability of blockchain applications in environmental sustainability projects, two examples are provided herein.

Operational Costs Analysis for ERC-20 Token Issuance

```

1 // Solidity smart contract for ERC-20 based Carbon Token
  issuance
2 pragma solidity ^0.8.0;
3 import "@openzeppelin/contracts/token/ERC20/ERC20.sol";
4
5 contract CarbonToken is ERC20 {
6     address public administrator;
7
8     constructor() ERC20("CarbonToken", "CTN") {
9         administrator = msg.sender; // Set the
          administrator as the contract deployer
10    }
11
12    // Function to issue new tokens
13    function issueTokens(address to, uint amount) public {
14        require(msg.sender == administrator, "Only the
          administrator can issue tokens.");
15        _mint(to, amount);}}

```

Transaction Cost in ETH = $55,000 \times 30 \text{ gwei} = 1,650,000 \text{ gwei} = 0.00165 \text{ ETH}$,
 Transaction Cost in USD = $0.00165 \text{ ETH} \times 2073.56 \approx 3.42 \text{ USD}$.

This cost analysis indicates that each token-issuance transaction incurs a fee of approximately \$3.42 USD based on the current ETH/USD value of 2,073.56USD on March 25, 2025. It is important to note that while the cost per transaction might be manageable, the cumulative cost for multiple transactions can become significant, suggesting that Ethereum's current gas fee structure may impact the scalability of deploying such a system for large-scale environmental projects. Deployment costs are beyond our current scope of analysis, but are acknowledged as a critical factor in the overall budget. It is worth mentioning that Layer 2 solutions have the potential to significantly reduce these costs. However, a detailed analysis of Layer 2 solutions falls outside the scope of this paper. The Layer 2 ecosystem is continuously evolving, with ongoing research, improvements, and the development of alternative solutions aimed at enhancing scalability and reducing transaction fees on the Ethereum network.

Operational Costs Analysis for Algorand Standard Assets Token Issuance

```

1 // Algorand smart contract for ASA based Carbon Credit
  issuance
2 # Python code to demonstrate creating an ASA on Algorand
3 from algosdk import algod
4 from algosdk.future.transaction import AssetConfigTxn
5 from algosdk.v2client import algod
6 # Connect to an Algorand node

```

```

7 algod_address = "https://testnet-algorand.api.purestake.io
  /ps2"
8 algod_token = "YourApiKey"
9 headers = {
10     "X-API-Key": algod_token,}
11 algod_client = algod.AlgodClient(algod_token,
    algod_address, headers)
12 # Account setup (ensure the account is funded and holds
    enough Algos for fees)
13 params = algod_client.suggested_params()
14 # Define the ASA parameters
15 asset_name = "CarbonCredit"
16 unit_name = "CCU"
17 total_supply = 1000000
18 decimals = 0
19 # Create the ASA
20 txn = AssetConfigTxn(
21     sender=creator_address,
22     sp=params,
23     total=total_supply,
24     default_frozen=False,
25     unit_name=unit_name,
26     asset_name=asset_name,
27     manager=creator_address,
28     reserve=creator_address,
29     freeze=creator_address,
30     clawback=creator_address,
31     url="http://example.com",
32     decimals=decimals)

```

ASA Creation Txn Cost (Algo) = 0.001 Algo,

ASA Creation Txn Cost (USD) = 0.001 Algo $\times$ 0.2056 USD = 0.0002056 USD.

This cost analysis shows that creating an ASA incurs a transaction fee of approximately 0.0002056USD, based on the current Algo/USD value of 0.20506USD on March 25, 2025. However, this calculation only includes the Algorand blockchain transaction fee. PureStake API fees, which vary based on the plan and number of API calls, are not included in this analysis. Algorand's fixed transaction fee structure significantly reduces transaction costs compared to networks with variable gas fees, such as Ethereum. This lower cost structure improves the feasibility of deploying Algorand-based systems for large-scale environmental projects without the burden of high cumulative transaction costs. Deployment costs (e.g., smart contract deployment, server costs, and development costs), although not covered here, are crucial for complete budgeting and planning.

On the whole, we offer a comparative analysis among the chosen blockchains. The tables provided present a comparison of blockchain platforms, emphasizing their appropriateness for carbon credit management applications.

Blockchain	Type	Consensus	Smart Contracts	Interoperability	Governance	Energy Efficiency
Ethereum	Public	PoW to PoS	Yes	Limited	Decentralized	Moderate
Hyperledger Fabric	Permissioned	Pluggable	Chaincode	Limited	Configurable	Depends on Setup
Algorand	Public	Pure PoS	Yes	Limited	On-chain Voting	High
Polkadot	Heterogeneous	Nominated PoS	Yes	Yes	Referenda	Depends on Parachains
Tezos	Liquid PoS	LPos	Yes	Yes	On-chain Governance	High
Avalanche	Permissioned/Public	Avalanche	Yes	Yes	On-chain Governance	High

Table 1. General Properties of Selected Blockchain Platforms

Blockchain	Transaction Speed	Finality	Transaction Cost
Ethereum	Moderate (15-45 seconds)	Probabilistic Finality	Variable, influenced by network congestion
Hyperledger Fabric	High (within seconds)	Depends on consensus algorithm	Low, suitable for enterprise solutions
Algorand	Very High (5 seconds)	Fast and Irreversible	Low, scalable and energy-efficient
Polkadot	High (within seconds)	Depends on relay chain and parachain consensus	Variable, influenced by relay chain and parachain mechanisms
Tezos	Moderate to High (15-30 seconds)	Depends on consensus	Adjustable through governance, generally moderate
Avalanche	Very High (few seconds)	Fast and Irreversible	Low, scalable and energy-efficient

Table 2. Txn Speed, Finality, and Txn Cost for Selected Blockchains

6 Discussion

6.1 Implications of Findings

The study suggests that blockchain has the potential to play a significant role in redefining carbon credits, moving them from primarily tradeable commodities to more essential product attributes. This change challenges traditional economics by making a product’s environmental impact a key market factor. Blockchain enhances transaction transparency and security, helping consumers make environmentally sustainable choices. By embedding carbon credits into product lifecycles, it merges ecological responsibility with economic activity, rewarding sustainability in markets. This method could affect ecological impact assessments and play a role in incremental changes in worldwide economic frameworks and consumer habits. Additionally, it could drive worldwide regulatory and corporate shifts, rethinking carbon credits’ role in sustainability.

6.2 Comparison with Traditional Methods

Traditional carbon credit management treats credits as separate, tradeable assets, often disconnecting them from actual environmental impacts, leading to inefficiencies. Conversely, a blockchain-based approach integrates carbon credits into product lifecycles, making carbon footprints a core part of product value and enhancing transparency. Blockchain’s decentralization has the potential to reduce the role of intermediaries, which may help in reducing corruption and improving the reliability of carbon credit tracking. Smart contracts automate and record transactions, boosting compliance and accountability. This approach

aligns with environmental goals, creating a transparent market that discourages manipulation and encourages sustainability, significantly improving over traditional methods and offering a framework for global environmental governance transformation.

7 Conclusion

This article discusses the need for a blockchain-based database in carbon footprint tracking and management. With carbon footprint being a fundamental product attribute, similar to price, it can be effectively managed using blockchain tools rather than traditional data silos. Future product attributes will likely require management through token economics and blockchain systems. "If scalability and regulatory challenges are addressed, the proposed system has the potential to significantly improve global carbon footprint management.

7.1 Summary of Key Findings

The main discoveries of this study highlight the transformative capacity of blockchain technology in not only increasing the transparency and security of carbon credit transactions but also embedding these credits into the core structure of product and service lifecycles, thereby significantly modifying market dynamics. Firstly, Tokenization and Intrinsic Integration of Carbon Credits: The study illustrates that embedding carbon credits as integral attributes of products ensures a clear connection between a product's environmental impact and its market value. This integration enhances transparency and compels stakeholders to prioritize sustainability, effectively connecting economic activities with environmental impact. Secondly, Enhanced Transparency and Trust: Due to its inherent nature, blockchain technology provides an unprecedented level of transparency and traceability in transactions. This visibility is crucial in building trust among consumers and regulators by offering an unchangeable record of carbon emissions and credits, thus supporting more informed decision-making and improving regulatory compliance. In addition, Decentralization of Carbon Credit Management: The decentralization intrinsic to blockchain technology challenges the traditional carbon credit markets, which have historically been plagued by inefficiencies and vulnerability to manipulation. By removing intermediaries, blockchain technology decreases administrative costs and the potential for fraud, resulting in a more efficient and fair carbon credit market. Furthermore, Automation through Smart Contracts: Smart contracts automate multiple aspects of carbon credit transactions, including issuance, transfer, and retirement, thus simplifying compliance, reducing transaction costs, and minimizing human error. This automation ensures that environmental goals are seamlessly integrated into business processes, enhancing operational efficiency and adherence to sustainability standards. The findings indicate that blockchain technology possesses the capacity to markedly advance the management of carbon credits, thereby contributing substantially to

the progression of global sustainability objectives. The study lays a strong foundation for further investigation into how these technologies can be improved and broadened to promote a sustainable economic framework that aligns with conservation efforts globally. Through an examination of various blockchain architectures, including *Ethereum*, *Hyperledger Fabric*, *Algorand*, *Polkadot*, *Tezos*, and *Avalanche*, this study explains the potential of these platforms in addressing one of the complex challenges of environmental sustainability. Firstly, *Ethereum's* Important Role and Constraints: *Ethereum* stands out in the realm of decentralized applications by offering smart contract features essential for creating secure and transparent carbon management systems. However, its scalability challenges and the unpredictable nature of transaction costs underline the need for continuous technological improvements to enhance its effectiveness in extensive applications. In addition, *Ethereum* displays a paradox in these contexts. The platform has shifted towards becoming a value storage for market participants rather than resolving practical issues, consequently causing increasing prices and transaction expenses. Therefore, in the short term, *Ethereum* might not be an efficient tool for solving real-world issues. Secondly, *Hyperledger Fabric's* Enterprise-Level Solutions: The permissioned blockchain framework of *Hyperledger Fabric* provides a tailor-made solution for state-specific carbon management, focusing on data privacy and scalability. This platform's modular design offers a versatile approach, especially fitting for complex organizational systems that require strict data oversight. Nonetheless, there is still a need for a worldwide solution. *Hyperledger Fabric* reduces data silos and supports comparisons at the corporate level but only up to the individual state scale, which falls short. Additionally, *Algorand's* Efficiency and Scalability: *Algorand's* pure proof-of-stake method is distinguished by its high processing capability and quick transaction finality, presenting an energy-efficient option for monitoring carbon footprints. Yet, the long-term adoption and community backing remain unpredictable. Moreover, *Polkadot* and *Tezos* - Specialized Innovations: *Polkadot's* focus on interoperability and *Tezos's* on-chain governance give blockchain applications distinctive perspectives on an international level. These platforms provide specialized functions that can boost collaborative efforts and community-focused endeavors in managing carbon footprints. Although *Polkadot* seems to be a feasible choice, it still faces fluctuations in transaction costs. In addition, *Avalanche's* Advanced Consensus Protocol: The *Avalanche* platform, with its cutting-edge consensus process, provides a promising solution for high throughput and swift finality needs in carbon footprint tracking. Its recent introduction to the market indicates the potential for significant impact, although its developing ecosystem should be considered. Among various blockchains, *Avalanche* appears to be the most suitable candidate for such an application. To conclude, a globally governed system with extremely high finality and very low transaction costs will be essential.

7.2 Future Research Directions

Future research endeavors should firstly emphasize tackling the scalability and adaptability challenges of blockchain architectures to facilitate their widespread adoption across various industrial sectors. Secondly, it is crucial to explore the integration of advanced artificial intelligence (AI) and machine learning (ML) algorithms with blockchain technology to enhance the prediction and optimization of the carbon footprints of products and services. Firstly, concerning the scalability of blockchain systems, there is a need for additional research to increase their ability to efficiently handle the extensive transaction data generated by global carbon credit markets. Such efforts might involve the development of new consensus algorithms or alterations to existing ones to boost transaction throughput without sacrificing security or decentralization. Secondly, regarding regulatory frameworks and standardization, it is important to establish comprehensive frameworks accommodating the decentralized nature and global reach of blockchain. Research should also aim to create standardized protocols for carbon credit verification and transfer to ensure consistency and interoperability across borders. In addition, investigating the potential of blockchain to facilitate the trading and distribution of renewable energy credits, alongside carbon credits, could significantly impact both markets. Studies may examine how blockchain could integrate these systems to establish a unified platform that encourages sustainable development. Furthermore, the application of AI and ML techniques for analyzing blockchain data can provide a deeper understanding of consumption patterns and assist in predicting future trends in carbon emissions. Such endeavors would refine carbon management strategies and improve predictive accuracy regarding environmental impacts. Additionally, examining the human and organizational behavior associated with adopting blockchain technology for carbon credit management can shed light on the barriers and facilitators related to its adoption. Research might focus on psychological, cultural, and institutional factors influencing the acceptance and utilization of blockchain systems within various industries. Finally, conducting longitudinal studies to evaluate the long-term environmental, economic, and social impacts of carbon credit systems integrated with blockchain is essential. Such studies can quantify the tangible benefits and potential issues of these systems over time, thereby providing a more comprehensive understanding of their sustainability. These research directions aim to refine technological applications and comprehend the effect of blockchain on social norms and environmental policies, which can contribute to the development of robust, transparent, and efficient systems for managing global resources and promoting sustainable development.

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A Appendix

A.1 Terminology

(1) Carbon footprint Refers to the total greenhouse gas emissions associated with a product or process, typically expressed in terms of carbon dioxide equivalents (CO₂e). **(2) Carbon Credits** Tradable units that represent a specific amount of carbon dioxide or equivalent greenhouse gas that has been reduced, avoided, or offset. Carbon credits are obtained through tokenization of carbon footprints. **(3) Tokenization** The process of representing assets, data, or rights as digital tokens on a blockchain, allowing secure and efficient management.